

RHCSA 9v EX200 — Combined Study Notes

Combined from the `RHCSA-9_EX200` repository (markdown + plain-text study notes). Every source file begins on its own page; use the table of contents below to jump around.

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Overview

This repository is a comprehensive resource for individuals preparing for the **Red Hat Certified System Administrator (RHCSA) 9v EX200** exam. It contains the latest exam dumps, lab handbooks, and solved exercises to help you master the skills required for certification.

Contents

1. Dumps

- Collection of the latest **RHCSA EX200** exam dumps.
- Practice questions curated to mirror the actual exam format.
- Designed to enhance your troubleshooting and command-line skills.

2. Lab Handbooks

- **SA1 (System Administration I)**: Covers the foundational concepts of Linux system administration, including basic tools, shell scripting, and user management.
- **SA2 (System Administration II)**: Focuses on advanced topics such as networking, storage, and security.

3. Solved Latest Dumps

- Detailed solutions to the latest RHCSA EX200 dumps.
 - Step-by-step explanations of each task to build confidence and understanding.
 - Helps in revising key concepts and commands effectively.
-

Lab Setup

To make the most out of this repository, it is recommended to set up a lab environment. Follow these steps:

1. Download and Install VMware Workstation

VMware Workstation is a powerful virtualization tool that allows you to run multiple operating systems on your computer. Download and install VMware Workstation from the following link:

📄 [Download VMware Workstation](#)

2. Download Red Hat Enterprise Linux (RHEL) 9 ISO

You will need the RHEL 9 64-bit ISO file to create your virtual machine. Follow these steps to download it:

Steps to Download RHEL 9 ISO

1. Create a Red Hat Account (if not already done):

- Visit the [Red Hat Login Page](#).
- Fill in your details and create an account.

2. Sign Up for Red Hat Developer Subscription:

- Go to the [Red Hat Developer Subscription Page](#).
- Log in with your Red Hat account.
- Click on "**Join now**" or "**Activate your subscription**" to get a free developer subscription.

3. Download the ISO:

- After activating your subscription, go to the [RHEL 9 Download Page](#).
- Select **RHEL 9 64-bit ISO** and download the file.

4. Verify the ISO Checksum (Optional but Recommended):

- Download the checksum file from the same page.

- Verify the ISO integrity using the following command:

```
sha256sum /path/to/downloaded.iso
```

- Compare the output with the checksum provided on the Red Hat website.
-

3. Set Up a Virtual Machine

- Open VMware Workstation.
 - Create a new virtual machine and select the RHEL 9 ISO file as the installation media.
 - Allocate at least **2 CPUs**, **4 GB RAM**, and **20 GB disk space** to the virtual machine.
 - Complete the installation process and set up a user account for practice.
-

How to Use This Repository

1. Dumps:

- Use the dumps to familiarize yourself with the exam pattern.
- Attempt the questions and compare your solutions with the solved dumps.

2. Lab Handbooks:

- Follow the SA1 and SA2 handbooks to learn and practice RHCSA topics.
- Use the provided labs to simulate real-world scenarios.

3. Solved Dumps:

- Review the solved dumps for answers and explanations to refine your skills.
 - Use them as a guide for troubleshooting and command execution.
-

Red Hat Certification Exam Guide

Exam Methods

1. **Kiosk** - Testing Center.
 2. **Remote** - Requirements:
 - Bootable USB drive.
 - Two cameras.
 - High-performance laptop.
 - Reliable internet connectivity.
 - Power backup.
-

Candidate Requirements

- **Original ID Card:**
 - Must be pre-approved by Red Hat during exam scheduling.
 - Accepted IDs:
 - Aadhar Card
 - PAN Card
 - Driving License
 - Passport
 - **Note:** Laminated IDs are **not** accepted.
-

Pre-Exam Steps

1. **Provide the Following Details:**
 - RHN ID.
 - Exam Code: EX200 .
 2. **Read Instructions Carefully:**
 - Duration: 10-15 minutes.
 - Includes:
 - Exam rules (do's and don'ts).
 - Node 1 and Node 2 IP addresses.
 - Credentials (e.g., usernames, passwords).
-

Exam Details

- **Total Questions:** ~22
 - **Passing Score:** 210
 - **Duration:** 3 hours
 - **Break Policy:** 10 minutes allowed after the first hour (included in the 3-hour total).
-

Exam Environment

- **Allowed Commands:**
 - `copy` command is functional.
 - `tab` autocompletion is available.
 - `man` pages can be used but frequent usage may impact the score.
 - **Question Handling:**
 - Mark questions as **Done** for review or revisit.
-

Performing the Exam

1. Access Virtual Machines (VMs):

- Click on **Activities**.
- Locate the four icons on the left side and select **VM Console**.
- This will display Node 1 and Node 2.
- Log in to both VMs as the **root user**.
- Use the terminal provided in the exam to SSH into the VMs.

2. Prerequisites for VM Connection:

- Root passwords (provided in instructions).
 - Ensure a working SSH connection.
-

Tasks Overview

1. Node 1:

- Configure the network:
 - Set a **static IP** to establish connectivity.

2. Node 2:

- Recover the **root password**:
 - Network configuration will already be in place.
-

Additional Notes

- During practice, VM names may be `servera` and `serverb`.
- In the exam, VM names could vary — adapt accordingly.

RHCSA v9 Exam Questions and Instructions

This document provides a comprehensive guide to RHCSA v9 exam tasks, including configuration and troubleshooting for two VMs:

- `node1.domainX.example.com` (IP: 172.24.10.10)
- `node2.domainX.example.com` (IP: 172.24.10.11)

Important Notes:

1. **Total Questions:** Approximately 22.
2. **VM Setup:**
 - One VM (`node2`) requires root password recovery.
 - Network, NTP, YUM, and other configurations are to be set up as per the instructions below.
3. **Prerequisites:**
 - SELinux and Firewall are enabled by default.
 - YUM repositories and other configurations must be implemented as described.

Tasks for `node1`

Q1: Configure Network with Static Parameters

Parameters:

- IP Address: `172.24.10.10`
- Netmask: `255.255.255.0`
- Gateway: `172.24.10.254`
- Nameserver: `172.24.10.254`
- Domain Name: `domainX.example.com`
- Hostname: `node1.domainX.example.com`

Steps:

```
ip a s
nmcli connection modify "Wired connection 1" \
    ipv4.address "172.24.10.10/24" \
    ipv4.dns "172.24.10.254" \
    ipv4.gateway "172.24.10.254" \
    ipv4.method manual
nmcli connection reload
hostnamectl set-hostname node1.domainX.example.com
```

Q2: Configure YUM Repositories

Repositories:

BaseOS: `http://content.example.com/rhel8.0/x86_64/dvd/BaseOS/`
AppStream: `http://content.example.com/rhel8.0/x86_64/dvd/AppStream/`

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RHCSA v9 Exam Instructions

Important Information

Please **read** the instructions carefully before starting.

Virtual Machines Details

```
1. **Node 1:**
  - Hostname: `node1.domainX.example.com`
  - IP Address: `172.24.10.10`
```

```
2. **Node 2:**
  - Hostname: `node2.domainX.example.com`
  - IP Address: `172.24.10.11`
```

Key Notes:

1. **Root Password:**
 - For one system, the root password is already **set** (no reset required).
 - For the other system, the root password needs to be recovered.
2. **Network Configuration:**
 - One system requires network configuration.
 - Networking is already configured on the other system.
3. **NTP Configuration:**
 - Required **for** **only one** system.
4. **YUM Repository:**
 - Must be configured on **both systems**.
5. **Firewall and SELinux:**
 - Both will be pre-enabled on all systems.

Exam Details

- **Total Questions:** Approximately 22
- **Scope of Tasks:**
 - Basic system administration tasks.
 - Troubleshooting and configuration.
 - Emphasis on critical services and security.

Best Practices:

- Ensure that configurations persist after reboots.
- Validate all configurations using provided tools or commands.
- Do not make unnecessary changes to pre-configured settings.
- Follow the given hostname and IP configurations strictly.

Good luck with your RHCSA v9 exam!

Node1 questions

- `node1` is running by default.

Q1. Configure network and set the static parameters.->consider machine configured as DHCP, need to config it with static parameters

Objective

Configure `node1` with static network parameters, update the hostname, and **enable** SSH connectivity.

Configuration Steps

Static Network Configuration

1. Set the following static parameters:

- **IP Address:** `172.24.10.10`
- **Netmask:** `255.255.255.0`
- **Gateway:** `172.24.10.254`
- **Nameserver:** `172.24.10.254`
- **Domain Name:** `domainX.example.com`
- **Hostname:** `node1.domainX.example.com`

2. Execute the following commands:

```
``bash
# ip a s
# cat /etc/resolv.conf
# ip route
# nmcli connection show
# nmcli connection modify "Wired connection 1" ipv4.address "172.24.10.10/24" \
    ipv4.dns "172.24.10.254" ipv4.gateway "172.24.10.254" ipv4.method manual
# nmcli connection reload
```

OR

```
nmcli connection modify ens160 ipv4.addresses 192.168.245.50/24
nmcli connection modify ens160 ipv4.gateway 192.168.245.1
nmcli connection modify ens160 ipv4.dns-search "domainX.example.com"
nmcli connection modify ens160 ipv4.dns "8.8.8.8 8.8.4.4"
nmcli connection modify ens160 ipv4.method manual
nmcli connection reload
nmcli connection up ens160
nmcli connection show ens160
# Verify the domain name in the resolv.conf
cat /etc/resolv.conf
#To Set back to DHCP --> nmcli connection modify ens160 ipv4.method auto
```

Hostname Configuration Verify and set the hostname:

```
# hostname
# hostnamectl set-hostname node1.domainX.example.com
```

SSH Configuration Edit the SSH configuration file:

```
# vim /etc/ssh/sshd_config
```

Update the following lines:

```
41 PermitRootLogin yes
66 PasswordAuthentication yes
```

Save and exit (:wq).

Restart the SSH service:

```
# systemctl restart sshd.service
```

Verification Test SSH connectivity from the exam-host to node1.domainX.example.com:

```
ssh root@node1.domainX.example.com
```

Notes Ensure all configurations persist after reboot. Validate connectivity and verify settings using the following commands:

```
ip a s
cat /etc/resolv.conf
ip route
nmcli connection show
```

Q2 Configure YUM repos with the given link (2 repos: 1st is BaseOS and 2nd is AppStream).

- **BaseOS:** `http://content.example.com/rhel9.0/x86_64/dvd/BaseOS`
- **AppStream:** `http://content.example.com/rhel9.0/x86_64/dvd/AppStream`

Answer:

Step 1: Edit the YUM Repository Configuration File

Run the following command to create or edit the repository configuration file:

```
# vim /etc/yum.repos.d/rhel.repo
```

Insert the following content into the file:

```
[BaseOS]
name=BaseOS
baseurl=http://content.example.com/rhel8.0/x86_64/dvd/BaseOS/
gpgcheck=0
enabled=1

[AppStream]
name=AppStream
baseurl=http://content.example.com/rhel8.0/x86_64/dvd/AppStream/
gpgcheck=0
enabled=1
```

- Save and exit the file by typing `:wq` then
- Verify Configuration

```
# yum list autofs
```

Q3. Debugging SELinux for Web Server on Non-Standard Port 82

Steps to Debug and Fix the Issue:

1. Check the status of `httpd` service:

```
yum list httpd
systemctl status httpd.service
Ensure httpd starts automatically at boot:
```

```
systemctl enable --now httpd.service
```

Verify the web server's content:

```
ls /var/www/html/
# Output:
file1 file2 file3
```

Check the configuration for listening on port 82:

```
cat /etc/httpd/conf/httpd.conf | grep -i "listen"
```

Allow port 82 through the firewall:

```
firewall-cmd --permanent --add-port=82/tcp
firewall-cmd --permanent --add-service=http
firewall-cmd --reload
firewall-cmd --list-all
vim /etc/httpd/conf/httpd.conf
Change --> Listen 80 to 82 port
```

Check SELinux settings for allowed HTTP ports:

```
semanage port -l | grep "http"
# Output:
http_port_t tcp 80, 81, 443, 488, 8008, 8009, 8443, 9000
```

Add port 82 to SELinux's allowed ports:

```
semanage port -a -t http_port_t -p tcp 82
```

Verify SELinux settings for port 82:

```
semanage port -l | grep "http"
#Output:
http_port_t tcp 82, 80, 81, 443, 488, 8008, 8009, 8443, 9000
```

To Apply the Changes don't forget restart the service

```
systemctl restart httpd
```

Test the server:

```
curl localhost:82/file1
```

Q4. Create User Accounts with Supplementary Group

1. Create a group named **"sysadms"**.
2. Create users:
 - **natasha** and **harry**: Assign them to the supplementary group **"sysadms"**.
 - **sarah**: Assign a non-interactive shell and ensure it's **not** a member of **"sysadms"**.
3. Set the password for all users to **"trootent"**.

```
# Create the group "sysadms"
groupadd sysadms

# Create users "natasha" and "harry" and assign them to the supplementary group "sysadms"
useradd -G sysadms natasha
useradd -G sysadms harry

# Create user "sarah" with a non-interactive shell and ensure it's not part of "sysadms"
useradd -s /sbin/nologin sarah

# Set passwords for all users
echo "trootent" | passwd --stdin harry
echo "trootent" | passwd --stdin natasha
echo "trootent" | passwd --stdin sarah

# Verify the group and user creation
tail /etc/group
tail /etc/passwd
```

Q5. Configure a cron job that runs every 2 minutes and executes:

- logger "EX200 in progress" as the user natasha.

Steps:

```
# Restart the cron service
systemctl restart crond.service

# Edit the crontab for user natasha
crontab -eu natasha

# Add the following cron job
*/2 * * * * logger "EX200 in progress"

# Save and exit (:wq)

# Restart the cron service again
systemctl restart crond.service
```

Verification:

```
#Check the cron job for user natasha:
crontab -l -u natasha
```

```
# Verify the logs to ensure the job is running:
tail -f /var/log/messages
```

Explanation of the Cron Expression:

```
*/2 * * * * logger "EX200 in progress"
```

***/2: Runs the job every 2 minutes. * (1st position): Every hour. * (2nd position): Every day of the month. * (3rd position): Every month. * (4th position): Every day of the week. logger "EX200 in progress": Logs the message "EX200 in progress" to the system logs.**

Q6. Create a Collaborative Directory

Task Description:

- Create the directory `/home/manager` with the following characteristics:
 1. Group ownership of `/home/manager` should go to the **"sysadms"** group.
 2. The directory should have full permissions for all members of the **"sysadms"** group, but no permissions for other users except **"root"**.
 3. Files created in the directory should automatically inherit the **"sysadms"** group ownership.

Steps:

```
# Step 1: Create the directory
mkdir /home/manager

# Step 2: Change the group ownership to "sysadms"
chgrp sysadms /home/manager
OR
chown :sysadms /home/manager

# Step 3: Verify the group ownership
ls -ld /home/manager
# Output:
# drwxr-xr-x 2 root sysadms 4096 <date> /home/manager

# Step 4: Set permissions for the directory with a setgid bit
chmod 2770 /home/manager
OR
chmod 770 /home/manager
chmod g+s /home/manager
# Step 5: Verify the permissions
ls -ld /home/manager
# Output:
# drwxrws--- 2 root sysadms 4096 <date> /home/manager

# Step 6: Create a test file in the directory
touch /home/manager/file1.txt

# Step 7: Verify the group ownership of the new file
ls -l /home/manager/file1.txt
```

```
# Output:
# -rw-r--r-- 1 root sysadms 0 <date> /home/manager/file1.txt
```

Q7. Configure NTP - Synchronize time of your system with the server 'utility.example.com'

Steps:

```
# Step 1: Check the current system time and NTP status
timedatectl
service ntp start
# Step 2: Restart the chrony service
systemctl restart chronyd.service

# Step 3: Edit the chrony configuration file
vim /etc/chrony.conf

# Step 4: Add the NTP server entry
server utility.example.com iburst

# Step 5: Save and exit the editor (:wq!)

# Step 6: Restart the chrony service to apply the changes
systemctl restart chronyd.service

# Step 7: Verify the system time and NTP synchronization
timedatectl

# Step 8: Check the NTP sources
chronyc sources -v
```

```
#Expected Outputs:
#After Step 7
timedatectl
Output:
Local time: <current_time>
Universal time: <current_utc_time>
RTC time: <rtc_time>
Time zone: <timezone>
NTP synchronized: yes
NTP service: active

#After Step 8 (chronyc sources -v):
Output:
210 Number of sources = 1
.utility.example.com          1  2  3  3  0.234  0.002  0.002
```

Q8. Configure AutoFS

Task Description:

- All remoteuserX home directories should be exported via NFS, which is available on utility.example.com (IP: 172.24.10.100).
- Your NFS exports directory is /home/remoteuserX for each remoteuserX .

- remoteuserX 's home directory should be automatically mounted using the AutoFS service.
- Home directories must be writable by their respective users.

Steps:

```
# yum install autofs -y

# Step 1: Enable and start the AutoFS service
systemctl enable autofs.service
systemctl start autofs.service

# Step 2: Configure AutoFS in the `auto.master` file
vim /etc/auto.master

# Add the following entry:
/rhome/remoteuserX /etc/auto.misc

:wq!

# Step 3: Configure AutoFS in the `auto.misc` file
vim /etc/auto.misc

# Add the following line:
remoteuserX -fstype=rw,nfs,soft,sync utility.example.com:/rhome/remoteuserX

:wq!

# Step 6: Restart the AutoFS service
systemctl restart autofs.service

# Verification:
# Switch to the `remoteuserX` account
su - remoteuserX

# Verify the mount point
pwd

# Should show:
/rhome/remoteuserX

# Test if the home directory is writable by creating a new file
$ touch file1.txt
```

```
su - remoteuserX
pwd
# Output should be:
/rhome/remoteuserX

Test by creating a new file:
$ touch file1.txt
```

Explanation:
Verify the existence of remoteuserX:

The `getent passwd remoteuserX` command checks if the user `remoteuserX` exists on the system.
Install AutoFS:

The `yum install autofs.x86_64 -y` command installs the AutoFS service, which will help in automatically mounting remote home directories when accessed.

Enable and start AutoFS:

The `systemctl enable autofs.service` ensures that the AutoFS service starts automatically at boot, and the `systemctl start autofs.service` starts the service immediately.

Configure AutoFS in `auto.master`:

The entry `/rhome/remoteuserX /etc/auto.misc` in the `auto.master` file tells AutoFS to use the `/etc/auto.misc` configuration file when mounting directories for `remoteuserX`.

Configure AutoFS in `auto.misc`:

The line `remoteuserX -fstype=rw,nfs,soft,sync utility.example.com:/rhome/remoteuserX` in the `auto.misc` file specifies the properties for mounting the remote user home directories over NFS.

Restart the AutoFS service:

Restarting the AutoFS service applies the new configurations.

Expected Outputs:

After Step 5, verify if the user home directory is mounted properly:

Q9. Create a Container Image from the Provided Link

Task Description:

- Create a container image from the **Containerfile** available at the link `http://utility.example.com/container/Containerfile`, and name the image as **process_files**.
- Login to the registry at `registry.lab.example.com` using the username **admin** and password **redhat321** (credentials available from the additional info page).

Steps:

```
# Step 1: Check the current user ID for athena
id athena

# Step 2: SSH into athena@localhost
ssh athena@localhost

# Step 3: Login to the container registry using podman
$ podman login -u admin
$ password: redhat321

# Step 4: Download the Containerfile from the provided link
$ wget http://utility.example.com/container/Containerfile

# Step 5: Build the container image using podman
$ podman build -t process_files -f .

# Step 6: List the images to verify the image creation
$ podman images
# Output should show:
# localhost/process_files
```

```
# Step 7: Exit the SSH session
$ exit
```

Expected Outputs:

After Step 6, when you run `podman images`, the output should look like:

plaintext

Copy code

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
localhost/process_files	latest	<image_id>	<timestamp>	<size>

###Explanation:

- **id athena:** The `id` command is used to check the user details for athena. This ensures that the user exists and the permissions are set correctly before proceeding.
- **ssh athena@localhost:** You SSH into the athena user account on the local machine to perform the necessary steps for building the container image.
- **podman login -u admin:** This command logs you into the container registry using the provided username (admin) and password (redhat321).
- **wget <http://utility.example.com/container/Containerfile>:** Downloads the Containerfile from the specified URL to the local machine.
- **podman build -t process_files -f .:** Builds the container image using the Containerfile and tags it as process_files.
- **podman images:** This command lists the local container images. You should see the newly created image localhost/process_files.
- **exit:** Exits the SSH session after completing the task.

Q10. Create Rootless Container and Do Volume Mapping

Task Description:

1. **Create a container named** `ascii2pdf` **using the previously created container image** `monitor`.
2. **Map** the local directories `/opt/files` and `/opt/processed` to the container directories `/opt/incoming` and `/opt/outgoing` respectively.
3. **Create a systemd service** called `container-ascii2pdf.service` to manage the container as a service.
4. **Enable the service** so it starts automatically after a reboot.

Steps:

```
# Step 1: Create and set up the local directories
mkdir /opt/files
chown -R athena:athena /opt/files

mkdir /opt/processed
chown -R athena:athena /opt/processed

# Step 2: SSH into athena user account
ssh athena@localhost
```

```

# Step 3: Run the container with volume mapping
$ podman run -d --name ascii2pdf -v /opt/files:/opt/incoming:Z -v /opt/processed:/opt/outgoing:Z localhost/monitor

# Step 4: Verify that the container is running
$ podman ps

# Step 5: Create the systemd user directory if not already present
$ mkdir -p /home/athena/.config/systemd/user/

# Step 6: Generate the systemd service for the container
$ cd /home/athena/.config/systemd/user/
$ podman generate systemd --name ascii2pdf --files --new

# Step 7: Verify the generated systemd service file
$ ls -lrt
# Output should show:
# container-ascii2pdf.service

# Step 8: Reload systemd user services
$ systemctl --user daemon-reload

# Step 9: Enable the systemd service to start on boot
$ systemctl --user enable container-ascii2pdf.service

# Step 10: Start the systemd service
$ systemctl --user start container-ascii2pdf.service

# Step 11: Enable lingering for the athena user to allow the service to run after logout
$ loginctl enable-linger athena

# Step 12: Verify user lingering is enabled
$ loginctl show-user athena

# Step 13: Verify the status of the service
$ systemctl --user stop container-ascii2pdf.service
$ podman ps
$ systemctl --user start container-ascii2pdf.service
$ podman ps

```

Expected Outputs:

After Step 4 (podman ps):

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
NAMES					
<container_id>	localhost/monitor	<command>	<timestamp>	Up <time>	
ascii2pdf					

After Step 12 (loginctl show-user athena):

User=athena

Linger=yes

After Step 13 (podman ps):

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
NAMES					
<container_id>	localhost/monitor	<command>	<timestamp>	Up <time>	
ascii2pdf					

Explanation:

Creating and setting up local directories:

- /opt/files and /opt/processed are created and ownership is set to the athena user. These directories will be mapped to the container's incoming and outgoing directories respectively. Running the container with volume mapping:
- The podman run command starts the ascii2pdf container with volume mappings (/opt/files:/opt/incoming and /opt/processed:/opt/outgoing). The :Z flag ensures that SELinux contexts are applied correctly. Generating a systemd service for the container:
- The podman generate systemd command generates a systemd service unit file for the container, which will manage the container lifecycle using systemctl. Enabling and starting the systemd service:
- The service is enabled using systemctl --user enable to ensure it starts on boot. It is then started using systemctl --user start. Enabling lingering for the athena user:
- The loginctl enable-linger athena command allows the athena user to have persistent services that continue running after they log out. Verification:
- After the service is started, systemctl --user stop and systemctl --user start verify that the service can be stopped and started successfully, while podman ps ensures the container is running.

Q11. Create User 'alex' with UID 3456 and Set the Password 'trootent'

Task Description:

- Create a user **alex** with the UID **3456**.
- Set the password for **alex** to **trootent**.

Steps:

```
# Step 1: Create the user 'alex' with UID 3456
useradd -u 3456 alex

# Step 2: Set the password for 'alex'
echo "trootent" | passwd --stdin alex

# Step 3: Verify the user creation and UID
id alex
Output: uid=3456(alex) gid=3456(alex) groups=3456(alex)
```

Q12. Locate All Files Owned by User "harry" and Copy Them Under /root/harry-files

Task Description:

- Locate all files owned by user **harry** and copy them to the directory **/root/harry-files**.

Steps:

```
# Step 1: Create the target directory for storing files
mkdir /root/harry-files

# Step 2: Find all files owned by user 'harry' and copy them to /root/harry-files
find / -user harry -exec cp -rvfp {} /root/harry-files \;

# Step 3: Verify the files copied
ls -a /root/harry-files
```

Q13. Find a String 'ich' from "/usr/share/dict/words" and Put It into /root/lines File

Task Description:

- Search for the string `ich` in the file `/usr/share/dict/words` and store the matching lines in the file `/root/lines`.

Steps:

```
# Step 1: Search for the string 'ich' in /usr/share/dict/words and save the result to /root/lines
grep "ich" /usr/share/dict/words > /root/lines

# Step 2: View the contents of /root/lines
cat /root/lines
```

Q14. Create an Archive '/root/backup.tar.bz2' of /usr/local Directory and Compress It with bzip2

Task Description:

- Create a compressed archive of the `/usr/local` directory and save it as `/root/backup.tar.bz2` using `bzip2` compression.

Steps:

```
# Step 1: Install bzip2 if not already installed
yum install bzip2.x86_64 -y

# Step 2: Create a compressed tar archive of /usr/local
tar cfvj /root/backup.tar.bz2 /usr/local
```

Explanation:

- `yum install bzip2.x86_64 -y`: Installs the `bzip2` package, which is required for compression.
- `tar cfvj /root/backup.tar.bz2 /usr/local`: The `tar` command creates a `.tar.bz2` archive:
- `c` - Create a new archive.
- `f` - Specifies the file name for the archive.
- `v` - Verbose output (optional, for displaying files being archived).

- j - Use bzip2 compression.
- The resulting archive will be stored at /root/backup.tar.bz2.

Expected Output:

- After running the tar command, a /root/backup.tar.bz2 file will be created containing the /usr/local directory.

Q15. Script to Store Search Result of Files in /usr/share Directory Greater Than 30k and Less Than 50k in /mnt/freespace/search.txt File

Task Description:

- Write a script that searches for all files in the `/usr/share` directory that are greater than 30k and less than 50k in size, and stores the search results in `/mnt/freespace/search.txt`.

Steps:

```
# Step 1: Create a new script file
vim test.sh

# Step 2: Write the script
#!/bin/bash
find /usr/share/ -uid 0 -size +30k -size -50k >/mnt/freespace/search.txt

# Step 3: Save and exit the editor (:wq!)

# Step 4: Make the script executable
chmod +x test.sh

# Step 5: Run the script
bash test.sh

# Step 6: Check the content of the search results
cat /mnt/freespace/search.txt
```

Explanation:

- vim test.sh: Opens the vim editor to create a new script file called test.sh.
- #!/bin/bash: This shebang line ensures that the script runs with the Bash shell.
- find /usr/share/ -uid 0 -size +30k -size -50k >/mnt/freespace/search.txt: The find command searches for files in the /usr/share directory that are:
 - Owned by root (-uid 0).
 - Between 30k and 50k in size (-size +30k -size -50k).
 - The results are redirected to /mnt/freespace/search.txt.
- chmod +x test.sh: Makes the script executable.
- bash test.sh: Executes the script.
- cat /mnt/freespace/search.txt: Displays the contents of the search.txt file, showing the list of matching files.

Expected Output:

- After running cat /mnt/freespace/search.txt, the output will display the list of files in /usr/share that are greater than 30k and less than 50k in size.

Node 2 Questions

Q16. Reset root user password and make it 'trootent'

Steps:

1. Reboot the System

- Press **Ctrl + Alt + Del** to reboot the system.

2. Select Rescue Mode

- When the GRUB menu appears:
 - Select **Rescue Mode** or **Troubleshooting Mode**.
- If **Rescue Mode** is not visible:
 - Highlight the default boot entry and press **e** to edit boot parameters.

3. Modify Kernel Parameters

- Locate the line starting with `linux` or `linux16`.
- Add the following at the end of the line:

```
rd.break
```

- Press **Ctrl + X** or **F10** to boot into the rescue shell.

4. Remount Root Filesystem

- In the rescue shell, remount the root filesystem in read-write mode:

```
mount -o remount,rw /sysroot
```

5. Access the Root Filesystem

- Change into the root filesystem:

```
chroot /sysroot
```

6. Reset the Root Password

- Set the new root password:

```
passwd
```

When prompted, set the password as trootent.

8. Trigger SELinux Relabeling

- Create a file to trigger SELinux relabeling during the next boot:

```
touch /.autorelabel
```

8. Exit and Reboot

- Exit the chroot environment:

```
exit
```

- Exit the rescue shell:

```
exit
```

- The system will reboot, and SELinux will relabel the filesystem.

9. Verify Root Access

- After rebooting, log in as the root user using the new password: Username: root Password: trootent

Important Notes:

SELinux Relabeling:

- The relabeling process may take some time, depending on the size of your filesystem.

Troubleshooting Mode:

- If a dedicated "Rescue" option is not visible, appending rd.break to the default boot entry achieves the same result.

Q17 Configure YUM repos with the given link (2 repos: 1st is BaseOS and 2nd is AppStream).

- **BaseOS:** `http://content.example.com/rhel9.0/x86_64/dvd/BaseOS`
- **AppStream:** `http://content.example.com/rhel9.0/x86_64/dvd/AppStream`

Answer:

Step 1: Edit the YUM Repository Configuration File

Run the following command to create or edit the repository configuration file:

```
# vim /etc/yum.repos.d/rhel.repo
```

Insert the following content into the file:

```
[BaseOS]
name=BaseOS
baseurl=http://content.example.com/rhel8.0/x86_64/dvd/BaseOS/
gpgcheck=0
enabled=1

[AppStream]
name=AppStream
baseurl=http://content.example.com/rhel8.0/x86_64/dvd/AppStream/
gpgcheck=0
enabled=1
```

- Save and exit the file by typing :wq then
- Verify Configuration

```
# yum list autofs
```

Q18. Resize a Logical Volume

Task Description:

Resize the logical volume **"mylv"** so that after reboot, its size should be between **290MB** and **330MB**.

Steps:

1. Check the Current Logical Volumes

- List all logical volumes:

```
lvs
```

2. Check the Filesystem Usage

- Display disk usage and filesystem type:

```
df -hT
```

3. Resize the Logical Volume

- Use `lvextend` to resize the logical volume **"mylv"** to 310MB:

```
lvextend -L 310M /dev/myvg/mylv
```

4. Resize the Filesystem

- After resizing the logical volume, resize the filesystem to match the new size:

```
resize2fs /dev/mapper/myvg-mylv
```

5. Verify the Logical Volume Size

- Check the logical volumes again to verify the new size:

```
lvs
```

6. Verify Disk Usage

- Verify the disk usage and the updated filesystem size:

```
df -hT
```

Q19. Add a Swap Partition of 512MB and Mount it Permanently

Task Description:

Add a swap partition of **512MB** and mount it permanently.

Steps:

1. Create the Swap Partition

- Use `gdisk` to create a new partition on `/dev/vdb` :

```
gdisk /dev/vdb
```

- Command (? for help): n to create a new partition
- Partition number (2-128, default 2): Press Enter

- First sector (34-2047, default = 34) or {+-}size{KMGTP}: Press Enter
- Last sector (34-2047, default = 2047) or {+-}size{KMGTP}: +512M
- Hex code or GUID (L to show codes, Enter = 8300): Press Enter

2. Set the Partition Type to Swap

- Change the partition type to Linux swap:

```
Command (? for help): t
Partition number (1-2): 2
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300): 8200
```

- Save and write the changes: Command (? for help): w

3. Settle Udev Rules and Reboot

- Settle udev changes:

```
udevadm settle
```

- Reboot the system to apply changes:

```
reboot
```

4. Verify the Partition

- Verify the new partition using lsblk:

```
lsblk
```

5. Create Swap Space

- Format the newly created partition as swap:

```
mkswap /dev/vdb2
```

6. Get the UUID of the Swap Partition

- Copy the UUID of the swap partition:

```
blkid /dev/vdb2
Example output:
UUID="e5a95dd4-0417-4229-a499-92b29fe9f201"
```

7. Edit /etc/fstab for Permanent Mount

- Open /etc/fstab in a text editor:

```
vim /etc/fstab
```

- Add the following line to automatically mount the swap partition on boot:

```
UUID=e5a95dd4-0417-4229-a499-92b29fe9f201 swap swap defaults 0 0
```

- Save and exit the editor (:wq!).

8. Mount All Filesystems

- Mount all filesystems defined in /etc/fstab:

```
mount -a
```

9. Enable Swap

- Activate the swap partition:

```
swapon /dev/vdb2
```

10. Verify the Swap

- Verify swap is active:

```
swapon -s
```

- Alternatively, you can use:

```
swapon -d
```

11. Check Memory Usage

Check the system's memory usage including swap: `bash free -m`

Q20. Create a Logical Volume of Name "newlv" from a Volume Group Name "newvg"

Task Description:

Create a logical volume named "**newlv**" from a volume group named "**newvg**" with physical extents of **16M** and the logical volume size of **50 extents**.

Steps:

1. Create a New Partition for LVM

- Use `gdisk` to create a new partition on `/dev/vdb` :

```
gdisk /dev/vdb
```

- Command (? for help): n to create a new partition
- Partition number (3-128, default 3): 3
- First sector (34-2047, default = 34) or {+-}size{KMGT}: Press Enter
- Last sector (34-2047, default = 2047) or {+-}size{KMGT}: +1024M
- Hex code or GUID (L to show codes, Enter = 8300): Press Enter

2. Set the Partition Type to Linux LVM

- Change the partition type to Linux LVM (code: 8e00):

```
Command (? for help): t
Partition number (1-3): `3`
```

```
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300): `8e00`
```

- Save the partition table:

```
Command (? for help): w
```

3. Settle Udev Rules

Settle udev changes:

```
udevadm settle
```

4. Create Physical Volume (PV)

- Create the physical volume using /dev/vdb3:

```
pvcreeate /dev/vdb3
```

5. Create Volume Group (VG)

- Create the volume group newvg with a physical extent size of 16M:

```
vgcreate -s 16M newvg /dev/vdb3
```

6. Verify Volume Group

- Verify the volume group creation:

```
vgdisplay
```

7. Create Logical Volume (LV)

- Create a logical volume named newlv with a size of 50 extents:

```
lvcreate -n newlv -l 50 newvg
```

8. Verify Logical Volume

- Verify the creation of the logical volume:

```
lvs
```

9. Format the Logical Volume

- Format the logical volume with the ext3 filesystem:

```
mkfs -t ext3 /dev/mapper/newvg-newlv
```

10. Get UUID of the Logical Volume

- Get the UUID of the new logical volume:

```
blkid
```

11. Create Mount Point

- Create a directory to mount the new logical volume:

```
mkdir /mnt/newlv
```

12. Edit /etc/fstab for Permanent Mount

- Open /etc/fstab in a text editor:

```
vim /etc/fstab
```

- Add the following line to /etc/fstab to mount the logical volume automatically on boot:

```
UUID=<UUID-of-newlv> /mnt/newlv ext3 defaults 0 0  
Replace <UUID-of-newlv> with the actual UUID obtained from blkid.  
Save and exit the editor (:wq!).
```

13. Mount All Filesystems

- Mount all filesystems defined in /etc/fstab:

```
mount -a
```

14. Verify Mount

Verify that the new logical volume is mounted:

```
df -hT
```

Q21. Configure System Tuning

Task Description:

Choose the recommended '**tuned**' profile for your system and set it as the default.

Steps:

1. List Available Tuned Packages

- Use the following command to list the available tuned packages:

```
yum list tuned
```

2. Restart the Tuned Service

- Restart the tuned service to ensure the latest changes take effect:

```
systemctl restart tuned.service
```

3. Check the Current Active Profile

- Check the current active profile:

```
tuned-adm active
Output:
sql
Copy code
Current active profile: balanced
```

4. Get the Recommended Profile

- Get the recommended profile for your system:

```
tuned-adm recommend
#Output:
virtual-guest
```

5. Set the Virtual-Guest Profile as Active

- Set the virtual-guest profile as the active profile:

```
tuned-adm profile virtual-guest
```

6. Verify the Active Profile

- Verify that the virtual-guest profile is now active:

```
tuned-adm active
Output:
- Current active profile: virtual-guest
```

Q1. Configure network and set the static parameters. (configure network in console)

- IP-ADDRESS= 172.24.10.10
- NETMASK= 255.255.255.0
- GATEWAY= 172.24.10.254
- (DNS) Nameserver= 172.24.10.254
- Domain Name= domainX.example.com
- hostname= node1.domainX.example.com

Q2. Configure YUM repos with the given link (2repos: 1st is BaseOS and 2nd is AppStream)

- BaseOS http://content.example.com/rhel8.0/x86_64/dvd/BaseOS
- AppStream http://content.example.com/rhel8.0/x86_64/dvd/AppStream

Q3. Debug SELinux - A web server running on non standard port 82 is having issues serving content, Debug and fix the issues.

- The web server on your system can server all the existing HTML files from /var/www/html
- Web service should automatically start at boottime.
- Do not make any changes to these files

Q4. Create User accounts with supplementary group.

- create the group a named "sysadms".
- create users as named "natasha" and "harry", will be the supplementary group "sysadms".
- cerate a user as named "sarah", should have non-interactive shell and it should be not the member of "sysadms".
- password for all users should be "trootent"

Q5. Configure a cron job that runs every 2minutes and executes: logger "EX200 in progress" as the user natasha.

Q6. Create a collaborative Directory.

- Create the Directory "/home/manager" with the following characteristics
- Group ownership of "/home/manager" should go to "sysadms" group
- The directory should have full permission for all members of "sysadms" group but not to the other users except "root"
- Files created in future under "/home/manager" should get the same group ownership

Q7. Configure NTP

- Synchronize time of your system with the server 'utility.example.com'

Q8. Configure AutoFS - All remoteuserX home directory is exported via NFS, which is available on utility.example.com(172.24.10.100) and your NFS-exports directory is /home/remoteuserX for remoteuserX

- remoteuserX's home directory is utility.example.com:/rhome/remoteuserX, where X is your station number and benath as /rhome/remoteuser5
- remoteuserX's home directory should be automounted autofs service.
- home directories must be writable by their users

Q9. Create a container image from the provided link.

- create a container image from "<http://utility.example.com/container/Containerfile>" name it as 'monitor'with user athena
- login to 'registry.lab.example.com' through "admin" and "redhat321" ->find it out credentials from Instructions page

Q10. Create rootless container and do volume mapping which they asked you in the question and run container as a service from normal user account, the service must be enable so it could start automatically after reboot

- Create a container named as 'ascii2pdf' using the previously created container image from previous question 'monitor'
- Map the '/opt/processed' to container '/opt/outgoing'
- Map the '/opt/files' to container '/opt/incoming'
- Create systemd service as container-ascii2pdf.service
- Make service active after all server reboots.

Q11. Create user 'alex' with 3456 uid and set the password 'trootent'

Q12. Locate all files owned by user "harry" and copy it under /root/harry-files

Q13. Find a string 'ich' from "/usr/share/dict/words" and put it into /root/lines file.

Q14. create an archive '/root/backup.tar.bz2' of /usr/local directory and compress it with bzip2

Q15. script. Store the search result of all files in the /usr/share directory that is greater than 30k and less than 50k in the /mnt/freespace/search.txt file

Node 2 ---- serverb Instructions

- Total they'll give you 2 disks with GPT Partitioning scheme (swap : 19 , LVM : 31 codes in GPT)
- /dev/vda (don't make any changes on /dev/vda)
- /dev/vdb (for swap and LVM)

Q15. Reset root user password and make it 'trootent' (users should answer in console only)

Q16. Configure YUM Repos

- BaseOS http://content.example.com/rhel8.0/x86_64/dvd/BaseOS
- AppStream http://content.example.com/rhel8.0/x86_64/dvd/AppStream

Q17. Resize a logical Volume -Resize the logical volume "mylv" so that after reboot size should be in between 290MB to 330MB

Q18. Add a swap partition of 512MB and mount it permanently

Q19. Create logical volume and mount it permanently

- Create a logical volume of name "wshare" from a volume group name "wgroup" physical extents of 16M and logical volume should have size of 50extents
- Mount logical volume with /mnt/wshare and format with ext3 filesystem

Q20. Configure System Tuning:

- Choose the recommended 'tuned' profile for your system and set it as the default

Extra Questions

```
fdisk /dev/vdb
command (m for help) : n
+1G
t
8E
w
partprobe
partprobe /dev/vdb
pvcreate /dev/vdb1
vgcreate myvg /dev/vdb1
lvcreate -n mylv -l 40 myvg /dev/vdb1
OR
lvcreate -n mylv -L 160M myvg /dev/vdb1
mkfs.ext4 /dev/myvg/mylv
mkdir /mylav
mount /dev/myvg/mylv /mylv
blkid
Copy --> /dev/mapper/myvg-mylv
vim /etc/fstab
/dev/mapper/myvg-mylv /mylv ext4 defaults 0 0
wq
mount -av
lsblk
```

Task

```
#lvs
Copy --> LV Path /dev/myvg/mylv
lvextend -L 260M /dev/myvg/mylv
lsblk
df -Th
resize2fs /dev/myvg/mylv
df -Th
```

Create 512MB SWAP Partition

```
fdisk /dev/vdb
n --> new
Last sector +512M
t
82
w
partprobe /dev/vdb
mkswap /dev/vdb2
UUID= swap swap defaults 0 0
mount -a
swapon /dev/vdb2
```

```
swapon -a
```

Create a logical volume of name "wshare" from a volume group name "wgroup" physical extends of 16M and logical volume should have size of 50 extents. Format with vfat filesystem and permanently mounted to /mnt/wshare

```
lvname = wshare
vgname = wgroup
```

```

physical extend 16M
lv lenght 50 extend
fdisk /dev/vdb
F
n
p
+1G
t
8E
w
partprobe /dev/vdb
pvcreate /dev/vdb4
vgcreate -s 16M wgroup /dev/vdb4
lvcreate -n wshare -l 50 wgroup /dev/vdb1
lvdisplay
copy --> LV Path : /dev/wgroup/wshare
mkfs.vfat /dev/wgroup/wshare
mkdir /mnt/wshare
blkid
Copy --> /dev/mapper/wgroup-wshare
vim /etc/fstab
/dev/mapper/wgroup-wshare /mnt/wshare vfat defaults 0 0

```

Q1. give **sudo** privillage to user or group

```

vim /etc/sudoers
add below
#Allow root to run any commands anywhere
root ALL=(ALL) ALL
username ALL=(ALL) ALL
username ALL=(ALL) NOPASSWD: ALL ---> Without password

```

For Group

```

## Allows people in group wheel to run all commnads
%wheel ALL=(ALL) ALL
%groupname ALL=(ALL) NOPASSWD: ALL

```

Q2.Set the default max and min days **for** the password policy.

```

vim /etc/login.defs
PASS_MAX_DAYS 90
PASS_MIN_DAYS 30
PASS_WARN_AGE 5

```

Q3. Set welcome note **for** the server.

```

vim /etc/profile
echo "WELCOME TO SERVERA"
source WELCOME TO SERVERA

```

Q4. Question on File Permission and **umask**

```

dr-xr-xr-x
-r--r--r--

```

Full Permission	777	666
Default	555	444
	222	222

```

useradd mohit

```

```
su - mohit
vim .bashrc
umask 222
source .bashrc
umask 222
touch f1
mkdir d1
ls -l
```

Q5. Script file

- create a script file with the name 'script.sh' inside the '/bin' and locate all files from '/usr/local' directory which has more than 30k and less than 50k copy under '/root/d1'.
- set the sgid permission for this directory.

```
mkdir -p /mnt/freespace
touch /mnt/freespace/search.txt
vim /bin/test.sh
```

```
#!/bin/bash
```

```
find /usr/share -size +30k -size -50k > /mnt/freespace/search.txt
chmod g+s /mnt/freespace
```

```
chmod +x test.sh
```

Q1. Give sudo privilege to user or group

To give `sudo` privileges to a user or a group, modify the `/etc/sudoers` file using `vim` (or another editor).

For a user:

```
# Allow root to run any commands anywhere
root ALL=(ALL) ALL

# Give sudo privileges to username
username ALL=(ALL) ALL

# Allow username to run commands without password
username ALL=(ALL) NOPASSWD: ALL
For a group:
# Allows people in group wheel to run all commands
%wheel ALL=(ALL) ALL

# Allow groupname to run commands without password
%groupname ALL=(ALL) NOPASSWD: ALL
- Make sure to use visudo to safely edit the /etc/sudoers file to avoid syntax errors.
```

Q2. Set the default max and min days for the password policy

- To set password expiration policies, modify the `/etc/login.defs` file.

```
# Edit /etc/login.defs:
vim /etc/login.defs
# Add or modify the following lines:
PASS_MAX_DAYS 90      # Maximum password age (90 days)
PASS_MIN_DAYS 30      # Minimum password age (30 days)
PASS_WARN_AGE 5       # Days before password expiration to warn user (5 days)
```

Q3. Set welcome note for the server

```
# To set a welcome message that appears when a user logs in, edit the /etc/profile file.
Edit /etc/profile:
vim /etc/profile

# Add the following line at the end of the file:
echo "WELCOME TO SERVERA"
Then, reload the profile to apply the changes:
source /etc/profile
```

Q4. Question on File Permission and umask

```
# File permissions and umask can be set using the following commands.
# Example Permissions:
# dr-xr-xr-x: Directory with read and execute permissions for everyone (octal 555).
# -r--r--r--: File with read-only permissions for everyone (octal 444).
# Full Permission (octal 777 for directories and 666 for files):

# Directories with full permissions
drwxrwxrwx
```

```
#Default Permissions:
# The default permissions for new files are determined by the umask value.
umask 222 # This gives directories with `555` permissions and files with `444` permissions
# Setting umask for a user:
useradd mohit
su - mohit
vim .bashrc

# Add the following line to .bashrc:
umask 222
# Then, apply the changes:

source .bashrc
Create a file and directory to observe the effect:
touch f1
mkdir d1
ls -l
```

Q5. Script File to Find Files and Set SGID Permission

- Steps to create a script and find files based on size:
- Create the script file script.sh inside /bin:

```
mkdir -p /mnt/freespace
touch /mnt/freespace/search.txt
vim /bin/script.sh
# Add the following script content:

#!/bin/bash

# Find files from /usr/local that are between 30k and 50k in size
find /usr/local -size +30k -size -50k > /mnt/freespace/search.txt

# Set SGID permission for the directory
chmod g+s /mnt/freespace

wq!

# Make the script executable:
chmod +x /bin/script.sh

# Explanation:
# find /usr/local -size +30k -size -50k: Finds files between 30k and 50k in size under /usr/local and
stores the results in search.txt.
# chmod g+s /mnt/freespace: Sets the SGID (Set Group ID) permission on the directory, ensuring that
files created within inherit the group ownership.
```

Configuration Settings:

`enabled=0` :

Disables the subscription-manager plugin, meaning YUM/DNF won't check for Red Hat subscriptions automatically.

Why do this?

To prevent automatic subscription checks or repository access, which may be unnecessary for non-Red Hat systems, or if you want to manually control subscriptions.

`disable_system_repos=0` :

Keeps third-party or custom repositories enabled, even when subscription-manager is triggered.

Why do this?

To allow access to non-Red Hat repositories while still using the system with subscription-manager for Red Hat updates. It ensures flexibility in managing custom or external repositories.

In Short:

These settings are used when you want to avoid automatic Red Hat subscription checks and keep non-Red Hat repositories active.

Resolve Subscription Issue

If you wish to disable the subscription-manager plugin and prevent automatic subscription checks, you can configure it as follows:

```
cd /etc/yum/pluginconf.d/  
vim subscription-manager.conf
```

Set the following line:

```
enabled=0  
- This will disable the subscription-manager plugin and stop automatic subscription checks during  
YUM/DNF operations.
```

Repo config

```
df -kh  
mount -a /run/media/servera/RHEL-9-0-0-BaseOS-x86_64  
cd /run/media/servera/RHEL-9-0-0-BaseOS-x86_64  
ll  
cd /etc/yum.repos.d/  
ll  
vim rhel.repo  
file:///run/media/servera/RHEL-9-0-0-BaseOS-x86_64/BaseOS  
file:///run/media/servera/RHEL-9-0-0-BaseOS-x86_64/BaseOS  
yum clean all  
yum repolist  
Verify
```

```
yum install httpd* -y
systemctl enable --now httpd
systemctl status httpd
```

Steps to Remount the Root Filesystem as Read-Write

Remount the Root Filesystem (/) as Read-Write: Since the root filesystem (/) is mounted as read-only, you can remount it as read-write using the following **command**:

```
sudo mount -o remount,rw /
```

This will remount the root filesystem **in** read-write mode, allowing you to make changes and run `lvcreate`.

Verify the Remount: After remounting, verify that the filesystem is now **in** read-write mode:

```
mount | grep "ro,"
```

The root filesystem (/dev/nvme0n1p3) should no longer appear with the `ro` flag.

```
fdisk -l
```

```
parted /dev/nvme0n1 print free
```

**Disk Management Basics ### **1. Understanding Partitioning** - **Partition:** Dividing a disk into separate sections for organizing data. - Types: - Primary: Up to 4 partitions. - Extended: Allows more partitions using logical volumes. ### **2. File Systems** - Common file systems in RHEL: ext4, xfs. - Default File System: xfs.**

Tasks and Commands

A. Creating Standard Partitions

1. View Disk Information:

```
df -Th
lsblk
fdisk -l
```

Create a Partition:

```
fdisk /dev/sdX
```

- n: Create a new partition.
- p: Primary or e for extended.
- Select sector range, type, and save with w. **Format the Partition:**

```
mkfs.xfs /dev/sdX1
```

- (Replace X1 with your disk name and partition number.)

Mount the Partition: `bash mkdir /mnt/newdisk mount /dev/sdX1 /mnt/newdisk` **Make the Mount Persistent (Edit /etc/fstab):** `bash /dev/sdX1 /mnt/newdisk xfs defaults 0 0`

B. SWAP Partition

Create a Partition:

```
fdisk /dev/sdX
```

- n: Create a new partition.
- p: Primary or e for extended.
- t: select file type
- L: List
- p: print for verify
- Select sector range, type, and save with w.

```
# Check Existing SWAP:
swapon --show
free -h
# Create SWAP Partition:
mkswap /dev/sdX2
swapon /dev/sdX2

# Add SWAP to /etc/fstab:
/dev/sdX2 swap swap defaults 0 0
```

C. Logical Volume Management (LVM)

- LVM: A flexible way to manage disk space, allowing resizing and snapshots.

Components:

- Physical Volume (PV): Physical disk or partition.
- Volume Group (VG): Collection of PVs.
- Logical Volume (LV): Allocatable units from VG.

Commands

```
# Create Physical Volume:
pvcreate /dev/sdX

# Create Volume Group:
vgcreate vg_name /dev/sdX

# Create Logical Volume:
lvcreate -L 10G -n lv_name vg_name

# Format and Mount LV:
mkfs.xfs /dev/vg_name/lv_name
mkdir /mnt/lv_mount
mount /dev/vg_name/lv_name /mnt/lv_mount

# Resize Logical Volume:
# Increase:
lvextend -L +5G /dev/vg_name/lv_name
xfs_growfs /mnt/lv_mount

# Decrease (requires unmounting):
umount /mnt/lv_mount
lvreduce -L -5G /dev/vg_name/lv_name
mount /dev/vg_name/lv_name /mnt/lv_mount
```

Conditions for Creating an LVM Hierarchy

1. Basic Conditions

- a. Physical Storage
 - **Requirement:** At least one physical storage device or partition.
 - Can be an unpartitioned disk (e.g., /dev/sdb) or a partition (e.g., /dev/sdb1).
 - The device should not already contain data unless you're repurposing it.
- b. Disk Label Type
 - The disk must have a valid label (MBR or GPT) created using parted or fdisk.
 - For LVM, the partition type should be marked as 8e (Linux LVM) when using fdisk.
- c. No Existing Mounts
 - Ensure the disk or partition is not already mounted or in use.
- Verify using:

```
lsblk
df -h
```

2. LVM Component Conditions

- **a. Physical Volume (PV)**
- The disk or partition must first be initialized as a Physical Volume.

```
pvccreate /dev/sdX
```

- **Conditions:**
- The disk must be writable.
- If repurposing a disk, clear existing partitions/data using wipefs:

```
wipefs -a /dev/sdX
```

- **b. Volume Group (VG)**
- At least one PV is required to create a Volume Group.

```
vgcreate vg_name /dev/sdX
```

- Conditions:
- A VG can consist of one or more PVs.
- The VG size equals the sum of the sizes of all PVs added to it.
- A VG must exist before creating an LV.

```
lvcreate -L 10G -n lv_name vg_name
```

- Conditions:
- The requested size (-L) must not exceed the free space in the VG.
- For Thin Provisioning, ensure sufficient space for the thin pool.

3. File System and Mounting

- a. Formatting After creating an LV, it needs a file system.

```
mkfs.xfs /dev/vg_name/lv_name
or
mkfs.ext4 /dev/vg_name/lv_name
```

- b. Mounting
- The LV can then be mounted to a directory:

```
mkdir /mnt/myvolume
mount /dev/vg_name/lv_name /mnt/myvolume
```

- 4. Practical Example of LVM Hierarchy Creation Step-by-Step

1. Create a Physical Volume (PV)

```
# 1. Identify the disk:
lsblk
# 2. Create PV:
pvccreate /dev/sdb
# 3. Create a Volume Group (VG)
# Add PV to VG:
vgcreate my_vg /dev/sdb
# 4. Create a Logical Volume (LV)
```

```
# Allocate 10GB for LV:
lvcreate -L 10G -n my_lv my_vg
# 5. Format and Mount LV
# Format the LV:
mkfs.xfs /dev/my_vg/my_lv
# Mount it:
mkdir /mnt/mydata
mount /dev/my_vg/my_lv /mnt/mydata
```

LVM Hierarchy Summary

Physical Volume (PV):

Base level representing raw disks or partitions.

pvcreate.

Volume Group (VG):

Groups multiple PVs into a single storage pool.

vgcreate.

Logical Volume (LV):

Allocates space from a VG for use as a file system or SWAP.

lvcreate.

Conditions for Advanced Operations

a. Extending an LV

Conditions:

Ensure the VG has free space.

Command:

```
# lvextend -L +5G /dev/vg_name/lv_name
```

```
# xfs_growfs /mnt/mountpoint
```

b. Reducing an LV

Conditions:

Unmount the LV before resizing.

Backup data as reducing can lead to data loss.

Command:

```
umount /mnt/mountpoint
```

```
# lvreduce -L -5G /dev/vg_name/lv_name
```

```
# mount /dev/vg_name/lv_name /mnt/mountpoint
```

Task 2: Set Up an LVM with 20GB Disk Space, Create a 10GB Logical Volume, and Expand It by 5GB

This document provides step-by-step instructions to configure a Logical Volume Manager (LVM) setup, create a 10GB Logical Volume (LV), and expand it by an additional 5GB.

Step 1: Identify Available Disks

Identify available unpartitioned disks to use for the LVM setup:

```
lsblk  
fdisk -l
```

Check the output for an unpartitioned disk (e.g., `/dev/sdc`).

Step 2: Create a Partition for LVM

1. Open the disk with `fdisk` :

```
fdisk /dev/sdc
```

2. Follow the prompts:

- Press `n` to create a new partition.
- Choose `p` for a primary partition.
- Accept the default partition number.
- Specify the starting sector (press Enter for default).
- Specify the ending sector (e.g., `+20G` for 20GB partition).
- Press `t` to change the partition type.
- Enter `8e` for the LVM partition type.
- Press `w` to write the changes and exit.

3. Inform the kernel of partition changes:

```
partprobe /dev/sdc
```

4. Verify the partition:

```
lsblk
```

You should see the new partition (e.g., `/dev/sdc1`).

Step 3: Create a Physical Volume (PV)

Convert the partition to a physical volume:

```
pvcreate /dev/sdc1
```

Verify the PV creation:

```
pvs
```

Step 4: Create a Volume Group (VG)

Create a volume group named `vgdata` using the physical volume:

```
vgcreate vgdata /dev/sdc1
```

Verify the VG creation:

```
vgs
```

Step 5: Create a Logical Volume (LV)

Create a 10GB logical volume named `lvdata` in the volume group `vgdata` :

```
lvcreate -L 10G -n lvdata vgdata
```

Verify the LV creation:

```
lvs
```

Step 6: Format and Mount the Logical Volume

1. Format the logical volume with the `ext4` file system:

```
mkfs.ext4 /dev/vgdata/lvdata
```

2. Create a mount point:

```
mkdir /mnt/lvdata
```

3. Mount the logical volume:

```
mount /dev/vgdata/lvdata /mnt/lvdata
```

4. Verify the mount:

```
df -h
```

Step 7: Expand the Logical Volume

1. Extend the logical volume by 5GB:

```
lvextend -L +5G /dev/vgdata/lvdata
```

2. Resize the file system to use the extended space:

```
resize2fs /dev/vgdata/lvdata
```

3. Verify the resized LV and file system:

```
lvs
df -h
```

Step 8: Configure Persistent Mount

To ensure the LV is mounted persistently after a reboot, add an entry to `/etc/fstab` :

1. Open `/etc/fstab` in a text editor:

```
nano /etc/fstab
```

2. Add the following line:

```
/dev/vgdata/lvdata    /mnt/lvdata    ext4    defaults    0 0
```

3. Save and exit the editor.

4. Test the configuration:

```
umount /mnt/lvdata
mount -a
```

5. Verify the mount:

```
df -h
```

Summary

You have successfully set up an LVM with 20GB disk space, created a 10GB logical volume, formatted and mounted it, and expanded the LV by 5GB. The logical volume is now ready for use and persistently mounted.

Check Available Free Space: Before running the `lvextend` command, you should verify how much free space is available in your volume group. You can use the following command to check:

```
vgdisplay
```

Look for the "Free PE" (Physical Extents) value to determine how much free space is available.

Extend the Logical Volume: If you are sure there is free space, you can run:

```
lvextend -l +100%FREE /dev/mapper/rhel-root
```

Resize the Filesystem: After extending the logical volume, you need to resize the filesystem to use the new space. Depending on the filesystem type, you can run one of these commands:

For ext4:

```
resize2fs /dev/mapper/rhel-root
```

For XFS (common in newer distributions):

```
xfs_growfs /dev/mapper/rhel-root
```

Verify the Changes: After completing the steps, you can check the new size of the logical volume and filesystem:

```
lvdisplay /dev/mapper/rhel-root
```

```
df -h
```

This will confirm that the logical volume has been extended and that the filesystem is now utilizing the additional space.

Task 4: Shrink an Existing LVM, Ensuring No Data Loss

This guide provides step-by-step instructions to safely shrink an existing Logical Volume (LV) while ensuring data integrity.

Prerequisites

1. Ensure you have a backup of critical data before resizing the LV.
 2. Verify the current usage of the file system to determine how much space can be safely freed.
-

Step 1: Verify the Current Logical Volume and File System Size

1. Check the logical volume details:

```
lvdisplay
```

2. Verify the file system size and usage:

```
df -h
```

3. Ensure that the used space in the file system is less than the target size.
-

Step 2: Unmount the Logical Volume

1. Unmount the logical volume:

```
umount /mount/point
```

2. Verify the unmounting:

```
df -h
```

Step 3: Run a File System Check

Before resizing, ensure the file system is consistent:

```
fsck -f /dev/vgname/lvname
```

Replace `vgname` and `lvname` with your volume group and logical volume names.

Step 4: Resize the File System

1. Shrink the file system to the desired size:

```
resize2fs /dev/vgname/lvname 10G
```

Replace `10G` with the target size of the file system.

2. Verify the new file system size:

```
resize2fs /dev/vgname/lvname
```

Step 5: Reduce the Logical Volume Size

1. Reduce the logical volume size to match the file system size:

```
lvreduce -L 10G /dev/vgname/lvname
```

Replace `10G` with the new logical volume size.

2. Confirm the reduction:

```
lvdisplay
```

Step 6: Remount the Logical Volume

1. Mount the logical volume back to its mount point:

```
mount /dev/vgname/lvname /mount/point
```

2. Verify the mount:

```
df -h
```

Step 7: Update `/etc/fstab` if Necessary

Ensure the mount configuration in `/etc/fstab` is correct if it was altered during the process:

1. Open `/etc/fstab` in a text editor:

```
nano /etc/fstab
```

2. Verify the entry for the logical volume.

3. Test the configuration:

```
umount /mount/point  
mount -a
```

4. Confirm the mount:

```
df -h
```

Summary

You have successfully shrunk an existing logical volume without data loss. The logical volume and its file system are now resized and remounted for use.

Task 5: Configure a Scenario Where Multiple Disks Are Added to a Volume Group

This guide explains how to add multiple disks to a Volume Group (VG) for Logical Volume Manager (LVM) configuration.

Step 1: Identify Available Disks

Use the following commands to check for available unpartitioned disks:

```
lsblk  
fdisk -l
```

Identify the disks to be added (e.g., `/dev/sdb` , `/dev/sdc`).

Step 2: Create Partitions for LVM

1. For each disk, create an LVM-compatible partition:

```
fdisk /dev/sdb
```

2. Follow the prompts:

- Press `n` to create a new partition.
- Press `p` for a primary partition.
- Accept the default partition number.
- Specify the starting sector (press Enter for default).
- Specify the ending sector (use the entire disk or required size).
- Press `t` to change the partition type.
- Enter `8e` for LVM.
- Press `w` to write the changes and exit.

Repeat the above steps for additional disks (e.g., `/dev/sdc`).

3. Inform the kernel of partition changes:

```
partprobe /dev/sdb  
partprobe /dev/sdc
```

4. Verify the partitions:

```
lsblk
```

Step 3: Create Physical Volumes (PVs)

Convert the partitions into physical volumes:

```
pvccreate /dev/sdb1 /dev/sdc1
```

Verify the PV creation:

```
pvs
```

Step 4: Create or Extend the Volume Group (VG)

1. If creating a new VG, run:

```
vgcreate vgdata /dev/sdb1 /dev/sdc1
```

Replace `vgdata` with your desired VG name.

2. If extending an existing VG:

```
vgextend vgdata /dev/sdb1 /dev/sdc1
```

3. Verify the VG:

```
vgs
```

Step 5: Create or Extend a Logical Volume (LV)

1. Create a new LV using the extended VG:

```
lvcreate -L 10G -n lvdata vgdata
```

Replace `10G` with the desired size and `lvdata` with the logical volume name.

2. If extending an existing LV:

```
lvextend -L +10G /dev/vgdata/lvdata
```

Replace `+10G` with the additional size.

3. Resize the file system to utilize the extended LV (if applicable):

```
resize2fs /dev/vgdata/lvdata
```

4. Verify the LV:

```
lvs
```

Step 6: Format and Mount the Logical Volume

1. Format the logical volume:

```
mkfs.ext4 /dev/vgdata/lvdata
```

2. Create a mount point:

```
mkdir /mnt/lvdata
```

3. Mount the logical volume:

```
mount /dev/vgdata/lvdata /mnt/lvdata
```

4. Verify the mount:

```
df -h
```

Step 7: Configure Persistent Mount

To make the mount persistent, add an entry to `/etc/fstab` :

1. Open `/etc/fstab` :

```
nano /etc/fstab
```

2. Add the following line:

```
/dev/vgdata/lvdata    /mnt/lvdata    ext4    defaults    0 0
```

3. Save and exit the editor.

4. Test the configuration:

```
umount /mnt/lvdata  
mount -a
```

5. Verify the mount:

```
df -h
```

Summary

You have successfully configured a scenario where multiple disks are added to a Volume Group. This setup allows flexible and scalable disk management for your logical volumes.

Task: Resize a Logical Volume to Be Between 290MB and 330MB

This guide explains how to resize an existing logical volume named `mylv` to a size within the range of 290MB to 330MB, ensuring persistence after a reboot.

Prerequisites

- Ensure you have sufficient free space in the volume group (`myvg`).
 - Verify the current size of the logical volume (`mylv`).
-

Step 1: Verify the Current Logical Volume and File System Size

1. Check the details of the logical volume:

```
lvs
```

2. Verify the file system type and usage:

```
df -hT
```

Step 2: Resize the Logical Volume

1. Resize the logical volume to 310MB:

```
lvextend -L 310M /dev/myvg/mylv
```

This command increases the size of `mylv` to 310MB.

2. Confirm the resizing:

```
lvs
```

Step 3: Resize the File System

1. Adjust the file system to utilize the new size:

```
resize2fs /dev/mapper/myvg-mylv
```

This ensures the file system uses the entire logical volume.

2. Verify the file system size:

```
df -hT
```

Step 4: Verify the Configuration Post-Reboot

To ensure the logical volume persists after a reboot:

1. Verify the entry in `/etc/fstab` :

```
nano /etc/fstab
```

Ensure the logical volume is correctly listed for automatic mounting.

2. Reboot the system:

```
reboot
```

3. After rebooting, confirm the size and mount status:

```
lvs  
df -hT
```

Commands Summary

```
# Check logical volume and file system details  
lvs  
df -hT  
  
# Extend the logical volume  
lvextend -L 310M /dev/myvg/mylv  
  
# Resize the file system  
resize2fs /dev/mapper/myvg-mylv  
  
# Verify the changes  
lvs  
df -hT  
  
# Verify after reboot  
reboot  
lvs  
df -hT
```

Summary

You have successfully resized the logical volume `mylv` to 310MB, ensuring that it is between 290MB and 330MB and persists after a reboot.

Add a 512MB Swap Partition and Mount it Permanently Using fdisk

Follow the steps below to add a swap partition of 512MB to your system using `fdisk` and mount it permanently.

1. Create the Swap Partition

```
# fdisk /dev/vdb
Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Command (m for help): p
Partition number (1-4, default 1): 2
First sector (2048-2047, default = 2048) or {+-}size{KMGTP}: Press Enter
Last sector, +sectors or +size{KMGTP}: +512M

Command (m for help): t
Partition number (1-4): 2
Hex code or GUID (L to show codes, Enter = 8300): 82

Command (m for help): w
```

This creates the swap partition and writes the changes.

2. Reboot the System

```
# udevadm settle
# reboot
```

3. Set Up the Swap Partition

After rebooting, run the following commands:

```
# lsblk
# mkswap /dev/vdb2
```

Copy the UUID for the swap partition (e.g., `e5a95dd4-0417-4229-a499-92b29fe9f201`), which you can get using:

```
# blkid /dev/vdb2
```

4. Edit `/etc/fstab`

Edit the `/etc/fstab` file to make the swap partition permanent by adding the UUID:

```
# vim /etc/fstab
UUID=e5a95dd4-0417-4229-a499-92b29fe9f201 swap swap defaults 0 0
:wq!
```

5. Mount the Swap Partition

Run the following commands to mount the swap partition and activate it:

```
# mount -a  
# swapon /dev/vdb2
```

6. Verify the Swap Partition

Check the swap status with:

```
# swapon -s  
# free -m
```

This will confirm that the swap partition is active and the system is using it.

Task 1: Create a New Partition of 5GB, Format It with ext4, and Mount It Persistently

This document outlines the steps to create a new partition of 5GB, format it with the `ext4` file system, and mount it persistently on a Linux system.

Step 1: Identify Available Disks

Use the following commands to identify available disks and ensure there is sufficient free space:

```
lsblk  
fdisk -l
```

Check the output for an unpartitioned disk (e.g., `/dev/sdb`).

Step 2: Create a 5GB Partition

1. Open the disk with `fdisk` :

```
fdisk /dev/sdb
```

2. Follow the prompts:

- Press `n` to create a new partition.
- Choose `t` to change the partition type (optional, if required).
- Select the partition type code, e.g., `83` for Linux.
- Press `p` to view the partition table.
- Specify the starting sector (press Enter for default).
- Specify the ending sector (e.g., `+5G` for a 5GB partition).
- Press `w` to write the changes and exit.

3. Inform the kernel of partition changes:

```
partprobe /dev/sdb  
fdisk -fp
```

4. Verify the partition:

```
lsblk
```

You should see the newly created partition (e.g., `/dev/sdb1`).

Step 3: Format the Partition

Format the new partition with the `ext4` file system:

```
mkfs.ext4 /dev/sdb1
```

Step 4: Create a Mount Point

Create a directory where the partition will be mounted:

```
mkdir /mnt/newpartition
```

Step 5: Mount the Partition Temporarily

Mount the partition to the directory:

```
mount /dev/sdb1 /mnt/newpartition
```

Verify the mount:

```
df -h
```

Step 6: Configure Persistent Mount

To make the mount persistent across reboots, add an entry to `/etc/fstab` :

1. Open `/etc/fstab` in a text editor:

```
nano /etc/fstab
```

2. Add the following line:

```
/dev/sdb1 /mnt/newpartition ext4 defaults 0 0
```

3. Save and exit the editor.

4. Test the configuration:

```
umount /mnt/newpartition  
mount -a
```

5. Verify the mount:

```
df -Th
```

Step 7: Validation

1. Check the partition size and usage:

```
lsblk
```

2. Verify the file system:

```
file -s /dev/sdb1
```

Summary

You have successfully created a 5GB partition, formatted it with `ext4` , and mounted it persistently. The partition will remain mounted even after a reboot.

Partition Plan

partition layout for a typical UEFI installation in GPT format, presented in a table:

Partition	Size	File System	Mount Point	Partition Type
/dev/nvme0n1p1	512MB	vfat (FAT32)	/boot/efi	EFI System Partition
/dev/nvme0n1p2	1GB	ext4 or xfs	/boot	Primary
/dev/nvme0n1p3	40GB	ext4 or xfs	/	Primary
/dev/nvme0n1p4	10GB	ext4 or xfs	/home	Primary
/dev/nvme0n1p5	4GB	ext4 or xfs	/var	Primary
/dev/nvme0n1p6	1GB	ext4 or xfs	/tmp	Primary
/dev/nvme0n1p7	4GB	swap	N/A	Primary (Swap)

Description of the Partitions:

EFI System Partition (/boot/efi):

Size: 512MB (minimum, can be larger if preferred)

File system: vfat (FAT32) – required for UEFI boot.

Partition Type: EFI System Partition.

Boot Partition (/boot):

Size: 1GB (usually enough for boot-related files).

File system: ext4 or xfs (you can choose based on preference).

Partition Type: Primary.

Root Partition (/):

Size: 40GB or more (depends on your needs).

File system: ext4 or xfs (most common choices).

Partition Type: Primary.

Home Partition (/home) (optional):

Size: 10GB or more (based on your usage needs).

File system: ext4 or xfs.

Partition Type: Primary.

Var Partition (/var) (optional):

Size: 4GB (this can vary based on the system's usage).

File system: ext4 or xfs.

Partition Type: Primary.

Temp Partition (/tmp) (optional):

Size: 1GB or more (depends on usage and requirements).

File system: ext4 or xfs.

Partition Type: Primary.

Swap Partition:

Size: 4GB or more (recommended based on the amount of RAM in the system).

File system: swap (used as virtual memory).

Partition Type: Primary (or Logical, if you have many partitions).

Notes:

EFI System Partition (/boot/efi) is crucial for UEFI boot, and it must be formatted with FAT32 (vfat). The root partition (/) can be the largest, typically around 40GB or more, depending on the system's purpose.

You may opt to create separate partitions for directories like /home, /var, and /tmp depending on your system needs.

Swap should generally be 4GB or more, but it's based on the available RAM and system usage.

This table ensures a proper partitioning scheme for a Red Hat 9 installation using UEFI and GPT. Let me know if you need further details!

Q9. Create a container image from the provided [link](#).

a. create a container image from

"<http://utility.example.com/container/Containerfile>" name it as 'process_files'

login to 'registry.lab.example.com' through "admin" and "redhat321" -> find it out credentials from additional info page

switch to user root

su - root

athena will be the existing user.

ssh athena@localhost

podman login registry.lab.example.com

username: admin

password: redhat321

-----> Download the Image <-----

wget <imagelink>

-----> Build the Image <-----

podman build -t <Image-Tag> -f .

podman build -t process_files -f .

-----> Check Podman Images

podman images

localhost/process_files (Output will get)

-----Extra

podman pull docker.io/centos/httpd

====+=====

Q10. Create rootless container and **do** volume mapping **which** they asked you **in** the question and run container as a service from normal user account, the service must be **enable** so it could start automatically after reboot.

a. Create a container named as 'ascii2pdf' using the previously created container image from previous question 'monitor'

b. Map the '/opt/files' to container '/opt/incoming'

c. Map the '/opt/processed' to container '/opt/outgoing'

d. Create systemd service as container-ascii2pdf.service

e. Make service active after all server reboots.

-----THIS has to **done** by root user -----

mkdir /opt/files

mkdir /opt/processed

ls -ld /opt/files

chown -R athena:athena /opt/files

chown -R athena:athena /opt/processed

-----This has to **done** by athena user -----

```
ssh athena@localhost
```

```
podman run -d --name ascii2pdf -v /opt/files:/opt/incoming:Z -v /opt/processed:/opt/outgoing:Z  
localhost/process_files
```

```
podman images
```

```
podman ps
```

```
mkdir -p .config/systemd/user
```

```
cd .config/systemd/user
```

```
podman generate systemd --name ascii2pdf --file --new
```

```
ls
```

```
systemctl --user daemon-reload      | Note: If you are facing issue --> loginctl enable-linger athena  
systemctl --user start container.ascii2pdf.service  
systemctl --user enable container.ascii2pdf.service  
systemctl --user status container.ascii2pdf.service
```

Q.22 **set** tuned recomonded profile

```
systemctl status tuned
```

```
yum install tuned.service
```

```
systemctl enable --now tuned
```

```
tuned-adm active
```

```
tuned-adm recomond
```

```
tuned-adm profile virtual-guest
```

```
tuned-adm
```

Steps to Configure Static IP Address

1. Open `nmtui` :

```
sudo nmtui
```

2. Navigate to **Edit Connection** and press `Enter` .
3. Select **Wired Connection 1** (or your desired connection) and choose **Edit**.
4. Update the following fields:
 - **IPv4 Configuration**: Set to `Manual` .
 - **Addresses**: Enter `172.25.250.10/24` .
 - **Gateway**: Enter `172.25.250.254` .
 - **DNS Server**: Enter `172.25.250.220` .
 - **Search domains**: Enter `lab.example.com` .
5. Enable the option **Ignore automatically obtained DNS Parameters**:
 - Highlight this option and press the `Space Bar` to select it.
6. Click **OK** to save the changes, then select **Back**.

Steps to Activate the Connection

1. Navigate to **Activate a Connection** and press `Enter` .
2. Perform the following actions:
 - Select **Wired Connection 1** (or your desired connection) and choose **Deactivate**.
 - Select **Wired Connection 1** again and choose **Activate**.
3. Confirm that the changes are applied.
4. Select **Back** to return to the main menu.

Steps to Set the System Hostname

1. From the main menu, navigate to **Set System Hostname** and press `Enter` .
2. Update the hostname:
 - Enter `servera.lab.example.com` as the new hostname.
3. Click **OK** to save the hostname and return to the main menu.
4. Exit `nmtui` .

Verification

1. Ping the configured IP address to verify the setup:

```
ping 172.25.250.10
```

- The ping should be successful, indicating that the static IP configuration is correct.

Configuring `dev` Group for Command Permissions on RHEL 8.6

This guide explains how to create the `dev` group on RHEL 8.6, add an existing user (`devuser`) to the group, and configure sudo permissions for specific commands.

Steps to Configure the `dev` Group

1. Create the `dev` Group

Run the following command to create the group:

```
sudo groupadd dev
```

- This creates a new group named `dev` .
- Verify the creation by running:

```
getent group dev
```

2. Add `devuser` to the `dev` Group

To add `devuser` to the newly created group, use the `gpasswd` command:

```
sudo gpasswd -a devuser dev
```

- Verify that `devuser` is now a member of the `dev` group:

```
groups devuser
```

3. Configure Sudo Permissions for the `dev` Group

Edit the `/etc/sudoers` file to define specific command permissions for the `dev` group.

(a) Open the Sudoers File Safely:

Use the `visudo` command to edit the file:

```
sudo visudo
```

(b) Add Permissions for the `dev` Group:

Append the following lines to the file:

```
%dev ALL=NOPASSWD: /bin/sh, /prd/utls/dev-GUI-stop.sh
%dev ALL=NOPASSWD: /bin/sh, /prd/utls/dev-gui.sh
%dev ALL=NOPASSWD: /bin/sh, /prd/utls/source_rate_automation_stop.sh
```

- These lines allow members of the `dev` group to execute the specified commands without entering a password.

(c) Save and Exit:

- Save and exit the editor (e.g., use `Ctrl+O` , `Enter` , then `Ctrl+X` in `nano` , or `:wq` in `vim`).

4. Verify the Configuration

(a) Check Group Membership:

Ensure that `devuser` belongs to the `dev` group:

```
groups devuser
```

(b) Test Command Execution:

Switch to `devuser` and test the commands:

```
sudo /bin/sh  
sudo /prd/utils/dev-GUI-stop.sh
```

These commands should execute without prompting for a password.

(c) List Allowed Commands for `devuser` :

Log in as `devuser` and check the allowed commands:

```
sudo -l
```

The output should display the commands permitted by the `dev` group configuration.

Additional Notes

- Always use `visudo` to edit the `/etc/sudoers` file to avoid syntax errors.
 - Ensure that the paths (`/prd/utils/...`) in the sudoers file are correct and accessible by `devuser` .
-